

Logical Structure of Oracle Database

Oracle logical storage structures

⇒ The datafiles in an Oracle database are grouped together into one or more tablespaces.

⇒ Within each tablespace, the logical db structures, such as tables and indexes, segments that are further subdivided into extents and blocks.

1. Tablespaces:

Def: A logical storage unit within an Oracle database.

- Consisting of one or more datafiles.

Datafile association:

- Each datafile belongs to one tablespace and cannot be shared with other tablespaces.

Minimum tablespaces in 11g:

- Two tablespaces:

SYSTEM Tablespace:

- Contains essential system metadata.

SYSAUX Tablespace:

- Contains auxiliary system data including data that was traditionally stored in the SYSTEM tablespace.

Default installations

- Creates six tablespaces including SYSTEM and SYSAUX.

Bigfile Tablespace:

Def:

- A tablespace with a single, large datafile that can be as large as 128TB.
- Simplifies the tablespace management.

OMF:

Def: A feature that automates the naming and management of datafiles, control files, and redo log files.

Benefits:

- Simplifies file management by allowing

the DBA to specify only the locations for these files.

Temporary Tablespace:

Def:

- A permanent tablespace used to hold temporary segments.

Usage:

- For sorting operations and temporary tables during user sessions, reducing I/O contention with permanent segments.

Tablespace management types:

Dictionary managed:

⇒ Extents are recorded in the data dictionary tables.

Locally managed:

⇒ Uses a bitmap in each datafile to track space usage.

⇒ Reduces contention for data dictionary tables.

SYSTEM Tablespace requirements:

⇒ If the SYSTEM tablespace is locally managed other tablespaces must also be locally managed for read and write operations.

⇒ Dictionary-managed tablespaces in such database must be read-only.

2. Blocks

Defn

⇒ The smallest unit of storage in an Oracle database.

Size

⇒ Defined by the DB_BLOCK_SIZE.

Multiplicity

⇒ Blocks can be of different sizes, but SYSTEM, SYSAUX, and temporary tablespaces must use the default DB_BLOCK_SIZE.

3. Extents

⇒ A logical unit of database storage made up of contiguous database blocks.

Usage

⇒ Allocated when a database object needs more space.

4. Segments

Defn

⇒ A collection of extents that make up of a database object.

Types of segments

Data segments - Every table in the database resides in a single data segment consisting of one or more extents.

• Stores table data, including LOB (large object) data segments for large binary objects. Oracle allocates more than one segment for a table if it is partitioned or clustered.

Index segments - Stores index data, including partitions for partitioned indexes and LOB index segments. Each index is stored in its own index segment.

Temporary segments

• Allocated for temporary operations like sorting that cannot be done in memory, existing only for duration of the operations of the SQL statement.

Rollback segments

⇒ Used to store old data values before changes, ensuring transactional integrity and read consistency.

Exist only in the SYSTEM tablespace and typically DBA does not need to maintain the rollback segments.

Automatic undo management

- Handles rollback segments in the undo tablespace.
- It automates the management of undo data, which stores old values before changes.
- Simplifies administration and improves performance.

Diagram:

Notes

- Already on previous pages.